

Resume AI coding work only from verified reality.

A checkpoint can preserve accepted decisions while repository facts become stale. HandoffGuard keeps the first and re-verifies the second.

accepted state survives

material facts re-verify

stale execution blocks

PASS	initial
PASS	unrelated
BLOCK	material drift
PASS	replan
FAIL	test
PASS	repair

The handoff problem is two different problems

Continuation state can remain valid while mutable repo facts become stale.

Preserve

- Accepted scope
- Accepted decisions
- Authority boundary
- Exact next move
- Lineage / transition identity

Re-verify

- Declared material paths + SHA256
- Acceptance command binding
- Source presence
- Optional Git HEAD context
- Final-current-byte proof

Restarting everything loses the left side. Blindly resuming trusts the right side.

Selective continuation

Repo changed $\neq$ material input changed. HandoffGuard distinguishes them before execution.



Unrelated repo change

Material hashes still match $\rightarrow$ PASS_RESUME_BINDING. Do not reopen accepted work.

Material change

Exact changed path reported $\rightarrow$ BLOCKED_STALE_HANDOFF before stale execution.

Executable proof sequence

The same acceptance command exposes the regression before the repair and proves the continuation after it.

01	Initial handoff	PASS_RESUME_BINDING
02	Unrelated HEAD change	PASS_RESUME_BINDING
03	Material source drift	BLOCKED_STALE_HANDOFF
04	Fresh-read + rebind	PASS_RESUME_BINDING
05	Unchanged acceptance	FAIL_EXECUTABLE_PROOF
06	Repair + rerun	PASS_EXECUTABLE_PROOF
07	Final current bytes	PASS_CONTINUATION

Local full-baseline proof: 14 tests PASS · deterministic sequence match · final adjudication PASS_CONTINUATION

Preserve accepted state. Re-verify mutable facts.

IBM Bob is in the material path

Bob does repository understanding, stale-state recovery, bounded repair and review.

1 Plan

Read sender handoff + current repo.
Identify accepted state vs mutable facts.
Explain why resume is blocked and which exact path needs fresh-read.

2 Agent

Verify stale handoff.
Fresh-read changed path.
Write bounded re-plan + successor handoff.
Run unchanged failing acceptance.
Repair only the proven defect.
Rerun proof.

3 Review

Review diff + lineage + receipts.
Check no silent scope widening.
Check screenshots and public-safe data.
Keep claims inside executable proof.

After each task: capture Bob task-session-summary → bob_sessions/

Full baseline first

A reduced variant may replace this only after it proves material net advantage with no material regression.

Full baseline keeps

- Accepted-state preservation
- Selective continuation
- Stale material-path fail-closed
- Fresh-read/re-plan lineage
- Executable acceptance
- Final-current-byte adjudication

Deliberately excluded

- Generic orchestration
- CI vendor integrations
- Backend/auth/database
- Private data
- Auto-publish
- Auto-submission

Claim ceiling

HandoffGuard proves a bounded continuation property.

It does not claim general correctness, production security, or autonomous authority.

python demo.py

One command reproduces the core story.